Laser Safety calculation for 940 nm LIDAR

Select exposure limit equation and correction factors from
EU Directive 2006/25/EC Annex II:

Table 2.5: Applied correction factors and other calculation parameters

Parameter as listed in ICNIRP	Valid spectral range (nm)	Value
C_A	$\lambda < 700$	$C_A = 1{,}0$
	700-1 050	$C_A = 10^{\,0.002(\lambda\,-\,700)}$
	1 050-1 400	$C_A = 5{,}0$
C_B	400-450	$C_B = 1{,}0$
	450-700	$C_B = 10^{\,0.02(\lambda\,-\,450)}$
C_C	700-1 150	$C_C = 1{,}0$
	1 150-1 200	$C_C = 10^{\,0{,}018(\lambda\,-\,1150)}$
	1 200-1 400	$C_C = 8{,}0$
T_1	$\lambda < 450$	$T_1 = 10$ s
	450-500	$T_1 = 10 \cdot [10^{\,0.02\,(\lambda\,-\,450)}]$ s
	$\lambda > 500$	T1 = 100 s
Parameter as listed in ICNIRP	Valid for biological effect	Value
α_{min}	all thermal effects	$\alpha_{min} = 1{,}5$ mrad
Parameter as listed in ICNIRP	Valid angular range (mrad)	Value
C_E	$\alpha < \alpha_{min}$	$C_E = 1{,}0$
	$\alpha_{min} < \alpha < 100$	$C_E = \alpha \,/\, \alpha_{min}$
	$\alpha > 100$	$C_E = \alpha^2 \,/\, (\alpha_{min} \cdot \alpha_{max})$ mrad with $\alpha_{max} = 100$ mrad
T_2	$\alpha < 1{,}5$	$T_2 = 10$ s
	$1{,}5 < \alpha < 100$	$T_2 = 10 \cdot [10^{\,(\alpha\,-\,1{,}5)\,/\,98{,}5}]$ s
	$\alpha > 100$	$T_2 = 100$ s
Parameter as listed in ICNIRP	Valid exposure time range (s)	Value
γ	$t \leq 100$	$\gamma = 11$ [mrad]
	$100 < t < 10^4$	$\gamma = 1{,}1\ t^{\,0.5}$ [mrad]
	$t > 10^4$	$\gamma = 110$ [mrad]

Table 2.2: Exposure limit values for laser exposure to the eye Short exposure duration < 10 s

Wavelength (λ) [nm]		Aperture	Duration [s]					
			10^{-15} - 10^{-11}	10^{-11} - 10^{-9}	10^{-9} - 10^{-7}	10^{-7} - $1{,}8 \cdot 10^{-5}$	$1{,}8 \cdot 10^{-5}$ - $5 \cdot 10^{-5}$	$5 \cdot 10^{-5}$ - 10^{-3}	10^{-3} - 10^{3}
UVC	180-280	1 mm for t < 0,3 s; $1{,}5 \cdot t^{0.375}$ for 0,3 < t < 10 s	$E = 3 \cdot 10^{10} \cdot$ [W m^{-2}] (a)		H = 30 [J m^{-2}]				
UVB	280-302				H = 40 [J m^{-2}]; if t < $2{,}6 \cdot 10^{-9}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	303				H = 60 [J m^{-2}]; if t < $1{,}3 \cdot 10^{-8}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	304				H = 100 [J m^{-2}]; if t < $1{,}0 \cdot 10^{-7}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	305				H = 160 [J m^{-2}]; if t < $6{,}7 \cdot 10^{-7}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$[J m^{-2}] (c)				
	306				H = 250 [J m^{-2}]; if t < $4{,}0 \cdot 10^{-6}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	307				H = 400 [J m^{-2}]; if t < $2{,}6 \cdot 10^{-5}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	308				H = 630 [J m^{-2}]; if t < $1{,}6 \cdot 10^{-4}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$[J m^{-2}] (c)				
	309				H = 10^{3} [J m^{-2}]; if t < $1{,}0 \cdot 10^{-3}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	310				H = $1{,}6 \cdot 10^{3}$ [J m^{-2}]; if t < $6{,}7 \cdot 10^{-3}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	311				H = $2{,}5 \cdot 10^{3}$ [J m^{-2}]; if t < $4{,}0 \cdot 10^{-2}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	312				H = $4{,}0 \cdot 10^{3}$ [J m^{-2}]; if t < $2{,}6 \cdot 10^{-1}$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	313				H = $6{,}3 \cdot 10^{3}$ [J m^{-2}]; if t < $1{,}6 \cdot 100$ then H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}] (c)				
	314				H = $5{,}6 \cdot 10^{3}\ t^{0.25}$ [J m^{-2}]				
UVA	315-400								
Visible & IRA	400-700	7 mm	H = $1{,}5 \cdot 10^{-4}\ C_E$ [J m^{-2}]	H = $2{,}7 \cdot 10^{4}\ t^{0.75}\ C_E$ [J m^{-2}]	H = $5 \cdot 10^{-3}\ C_E$ [J m^{-2}]		H = $18 \cdot t^{0.75}\ C_E$ [J m^{-2}]		
	700-1 050		H = $1{,}5 \cdot 10^{-4}\ C_A\ C_E$ [J m^{-2}]	H = $2{,}7 \cdot 10^{4}\ t^{0.75}\ C_A\ C_E$ [J m^{-2}]	H = $5 \cdot 10^{-3}\ C_A\ C_E$ [J m^{-2}]		H = $18 \cdot t^{0.75}\ C_A\ C_E$ [J m^{-2}]		
	1 050-1 400		H = $1{,}5 \cdot 10^{-3}\ C_C\ C_E$ [J m^{-2}]	H =$2{,}7 \cdot 10^{5}\ t^{0.75}\ C_C\ C_E$ [J m^{-2}]	H = $5 \cdot 10^{-2}\ C_C\ C_E$ [J m^{-2}]		H = $90 \cdot t^{0.75}\ C_C\ C_E$ [J m^{-2}]		
IRB & IRC	1 400-1 500	(d)	$E = 10^{12}$ [W m^{-2}] (b)		H = 10^{3} [J m^{-2}]			H=$5{,}6 \cdot 10^{3} \cdot t^{0.25}$ [J m^{-2}]	
	1 500-1 800		$E = 10^{13}$ [W m^{-2}] (b)		H = 10^{4} [J m^{-2}]				
	1 800-2 600		$E = 10^{12}$ [W m^{-2}] (b)		H = 10^{3} [J m^{-2}]			H=$5{,}6 \cdot 10^{3} \cdot t^{0.25}$ [J m^{-2}]	
	2 600-10^{6}		$E = 10^{11}$ [W m^{-2}] (b)		H=100 [J m^{-2}]	H = $5{,}6 \cdot 10^{3} \cdot t^{0.25}$ [J m^{-2}]			

We are using $\lambda = 940$ nm radiation at $t = 1.0$ s exposure time

$$C_A = 10^{0.002 \cdot (\lambda - 700)} = 3.02$$

We conservatively assume that our beam is perfectly collimated, therefore $\alpha < 1.5\ mrad$:

$$C_E = 1.0$$

Which gives us a maximum permissible exposure of:

$$H = 18 \cdot t^{0.75} \cdot C_A \cdot C_E \ \ \mathrm{Jm^{-2}} = 5.44\ mJ \cdot cm^{-2}$$